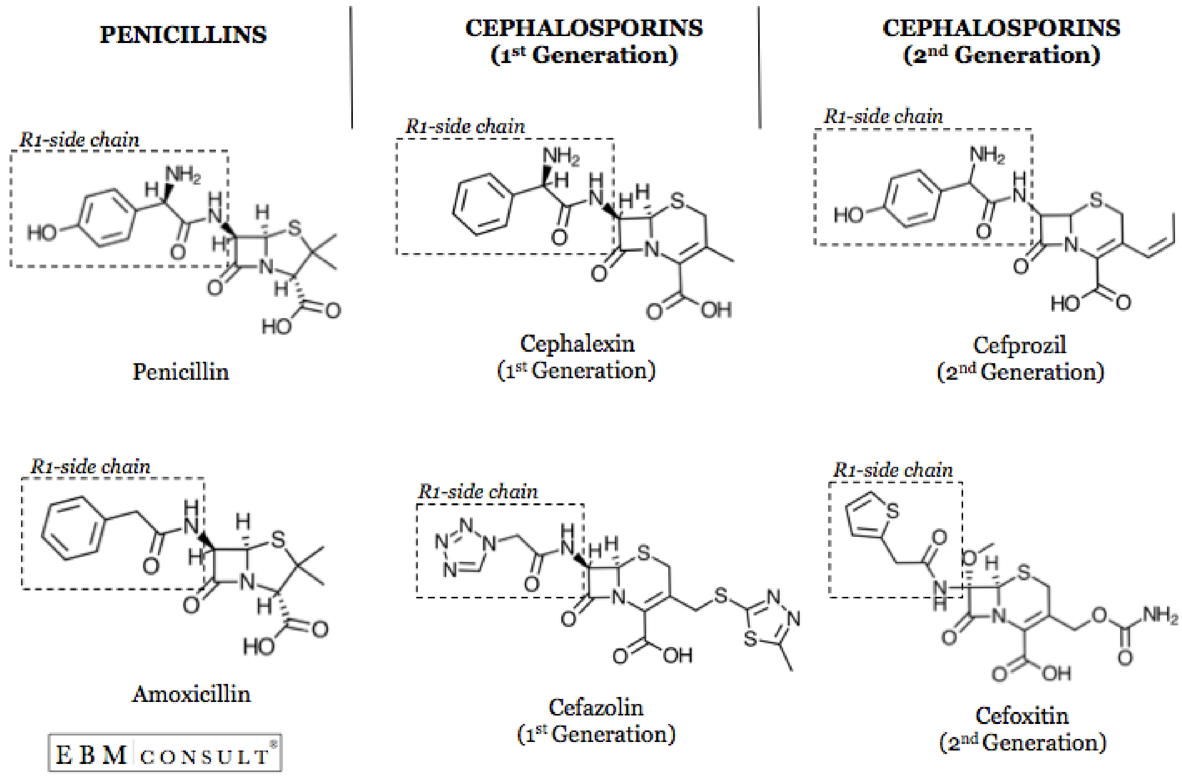
Cefazolin for Perioperative Patients with Penicillin Allergy

SITUATION

There is a shortage of intravenous clindamycin and much of the supply is utilized for perioperative antibiotic prophylaxis in patients that have a documented penicillin allergy.

BACKGROUND

Type-I hypersensitivity reactions mediated by IgE can cause urticaria, angioedema, bronchospasm, pruritis, or anaphylaxis within minutes to hours after medication administration. The similarity in structure of the R1-side-chains of penicillin antibiotics and cephalosporins is associated with the likelihood of cross-reactivity, not the presence of the beta-lactam ring. Cefazolin (Ancef), an intravenous first-generation cephalosporin, does not share a similar R1-side-chain with any of the current FDA-approved beta-lactam antibiotics.



Studies have evaluated the safety and efficacy of cefazolin for perioperative antibiotic prophylaxis in patients with penicillin allergy, even in cases of documented anaphylaxis:

Safety of Administering Cefazolin versus Other Antibiotics in Penicillin-Allergic Patients for Surgical Prophylaxis at a Major Canadian Teaching Hospital. Surgery. 2021 Sep. PMID: 33894984.	
Background	Cefazolin surgical prophylaxis is associated with better patient outcomes; however, its use in penicillin-allergic patients is controversial.
Methods	Quality improvement, retrospective chart review that assessed cefazolin prophylaxis among patients with an allergy to penicillin, including patients with a history of anaphylaxis. Antibiotic prophylaxis was classified as “cefazolin,” “clindamycin,” “vancomycin,” or “others.” Excluded patients with a history of delayed reactions such as Stevens-Johnson syndrome. Primary outcomes were adverse events in patients that received cefazolin perioperatively.
Results	27,467 operations were reviewed with 220 patients having a history of anaphylaxis with penicillin. 77 cefazolin orders, 63 clindamycin orders, 33 vancomycin orders, and 47 other antibiotic orders were placed. No statistical differences in allergic reactions (P = 0.70), surgical site infections (P = 1.00), or adverse events (P = 0.32) between cefazolin and other antibiotics. Clindamycin use decreased by 62% after implementation of policy that preferred cefazolin over clindamycin in this patient population. No cases of anaphylaxis were reported among patients that received cefazolin.
Conclusion	Administration of cefazolin in patients with an anaphylactic penicillin allergy for surgical prophylaxis appears to be safe.

Impact of the Allergy Clarification for Cefazolin Evidence-based Prescribing Tool on Receipt of Preferred Perioperative Prophylaxis: An Interrupted Time Series Study. Clin Infect Dis. 2020 Dec. PMID: 32364587.	
Background	Reported beta-lactam allergy is associated with increased risk of surgical site infection that is mediated by avoidance of cefazolin perioperative prophylaxis. Most patients who report beta-lactam allergy may be able to tolerate cefazolin given recent evidence that supports the lack of cross-reactivity between cefazolin and other beta-lactams.
Methods	Developed and utilized the Allergy Clarification for Cefazolin Evidence-based Prescribing Tool (ACCEPT) to screen for patients that could safely receive cefazolin (Figure I). Primary outcome was the monthly proportion of patients with reported beta-lactam allergy who underwent elective surgery and received cefazolin as perioperative prophylaxis. The secondary outcome was the number of perioperative allergic reactions.
Results	745 patients who reported beta-lactam allergy underwent 789 elective procedures where perioperative antibiotics were administered (48 patients had a history of angioedema or anaphylaxis to beta-lactams). At baseline, 40% of patients with a documented beta-lactam allergy received cefazolin as

	perioperative prophylaxis, compared to >80% after implementation of the ACCEPT. There were 2 instances of perioperative allergic reactions in the intervention period, more likely associated with the use of blue dye that happened to be used in both cases.
Conclusion	Implementation of a standardized tool for allergy assessment and perioperative antibiotic selection in patients with reported beta-lactam allergy resulted in a pronounced and sustained increase in the use of cefazolin.

A Streamlined Approach to Optimize Perioperative Antibiotic Prophylaxis in the Setting of Penicillin Allergy Labels. J Allergy Clin Immunol Pract. 2020 Apr. PMID: 31891825.	
Background	Penicillin-allergic patients often receive alternative antibiotics for perioperative prophylaxis, as opposed to first-line cephalosporins like cefazolin.
Methods	Developed institutional algorithm to follow (Figure II). Percent of patients receiving first-line antibiotics (cefazolin or cefuroxime) was assessed before and after implementation. Safety was assessed via chart review of patients who received epinephrine or diphenhydramine in the OR, or diphenhydramine within 24 hours postop.
Results	At baseline 22% of penicillin-allergic patients received first-line antibiotics, but after implementation >80% received first-line antibiotics (P <0.0001). Among 551 patients with penicillin allergy who received a cephalosporin, no reactions requiring epinephrine were noted. One patient had delayed rash that did not require discontinuation, and 3 patients received diphenhydramine for itching without rash with concomitant narcotic administration.
Conclusion	Using the algorithm, rates of first-line antibiotic administration for surgical prophylaxis significantly increased without severe adverse reactions.

The Impact of a Reported Penicillin Allergy on Surgical Site Infection Risk. Clin Infect Dis. 2018 Jan. PMID: 29361015.	
Background	A documented history of penicillin allergy may prevent administration of first-line antibiotic surgical prophylaxis to prevent surgical site infection (SSI).
Methods	Retrospective study that evaluated clinical outcomes in patients with and without penicillin allergies that underwent surgical intervention between 2010 and 2014. Primary outcome was occurrence of SSI.
Results	Evaluated 8,385 patients who underwent 9004 procedures. Patients with a documented penicillin allergy accounted for 11% of the population and 2.7% of patients had a SSI. Patients with a penicillin allergy had increased odds of developing a SSI (OR, 1.51; 95% CI, 1.02-2.22). These patients were also significantly more likely to receive clindamycin (P <0.001), vancomycin (P <0.001), and gentamicin (P <0.001) versus cefazolin (P <0.001).
Conclusion	Patients with a reported penicillin allergy were at increased odds of developing a SSI after receiving second-line perioperative antibiotics.

ASSESSMENT

Cefazolin is a safe option for antibiotic prophylaxis in patients that have a documented penicillin allergy even if the documented reaction is anaphylaxis; however, patients with a documented delayed hypersensitivity reaction such as SJS, TEN, AGEP, or DRESS should not receive cefazolin or other beta-lactam antibiotics.

RECOMMENDATION

Pending all other patient assessment considerations for use of cefazolin, screen patients with a documented penicillin allergy for appropriateness of cefazolin administration for perioperative antibiotic prophylaxis to limit unnecessary utilization of intravenous clindamycin. In the event that a patient has a recent history of MRSA infection(s) or is known to be colonized with MRSA, IV vancomycin should be considered depending on the surgery being done per the Clinical Practice Guidelines for Antimicrobial Prophylaxis in Surgery. Please contact Ronald Kendall, PharmD (x55829) or the ID fellow (UM pager 22789) for assistance with perioperative antibiotics.

Figure I:

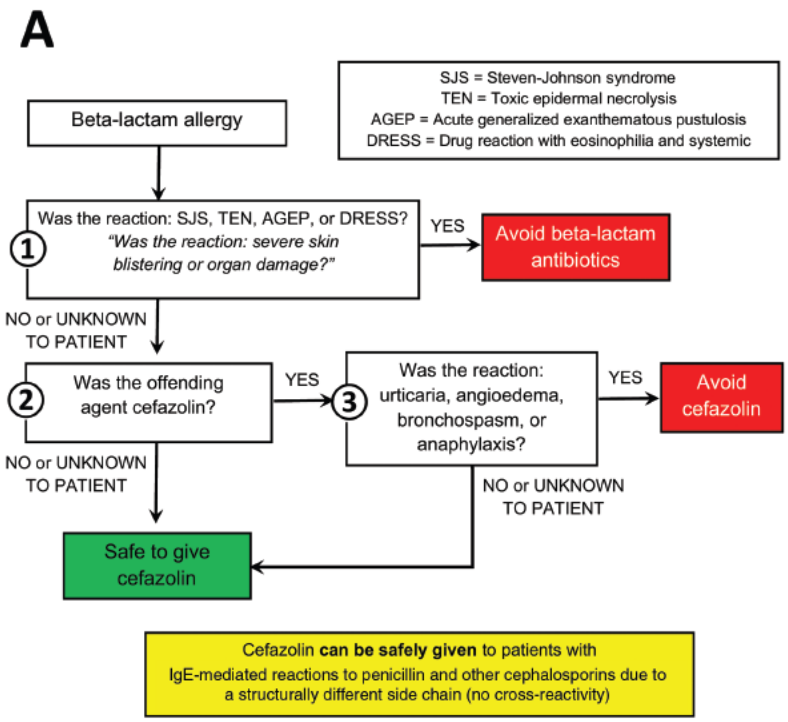
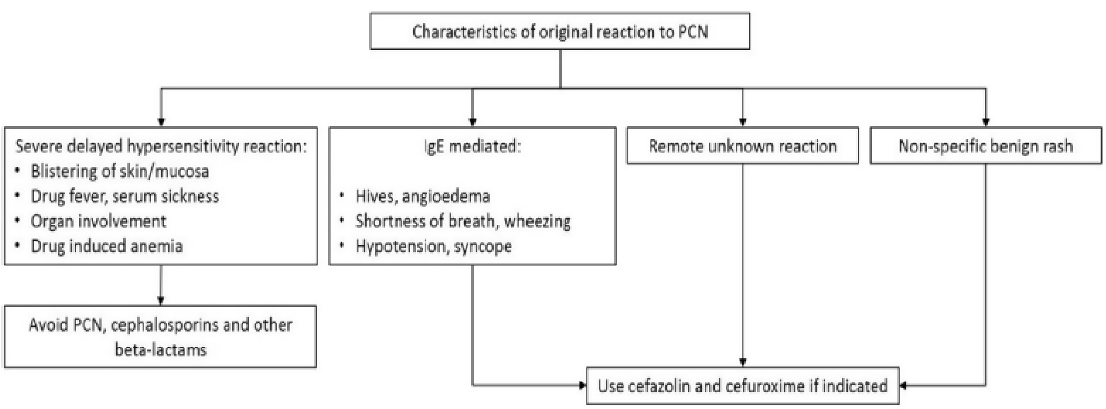


Figure II:



ADDITIONAL REFERENCES

- Penicillin and Cephalosporin Cross-Reactivity and Risk for Allergic Reaction (ebmconsult.com)
- Chaudhry SB, Veve MP, Wagner JL. Cephalosporins: A Focus on Side Chains and β-Lactam Cross-Reactivity. Pharmacy (Basel). 2019 Jul 29;7(3):103. doi: 10.3390/pharmacy7030103.
- Clinical Practice Guidelines for Antimicrobial Prophylaxis in Surgery (ashp.org)